

The area and perimeter of compound figures

Cut the shape into pieces you know, add or subtract, and trace the boundary separately.

LESSON 103–105

3 × 40 MIN

MATEMATIKA 6, §23

S7 15

LESSON CLOCK

25'

30'

35'

25'

5

Cutting into pieces	25 min
Subtracting	30 min
Perimeter by tracing	35 min
Mixed problems	25 min
Homework	5 min

LEARNING OBJECTIVES

- Split a compound figure into rectangles, triangles and parts of circles.
- Find an area by adding or by subtracting.
- Find the perimeter by tracing the boundary.
- Keep area and perimeter apart in the same question.

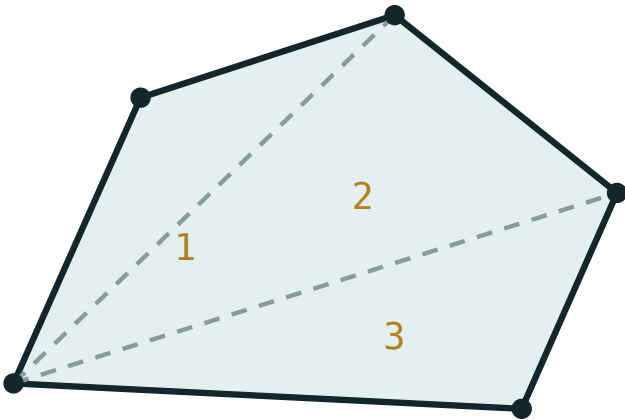
TEXTBOOK

National	pp. 301–310
Cambridge	Stage 7, pp. 148–156
Workbook	Exercise 15.4

01

Explanation

Cutting into pieces



split into triangles, then add the areas

A compound shape cut into known pieces

SHAPE	PIECES	AREA
an L-shape 10×8 with a 4×3 corner removed	two rectangles	68
a house outline: 6×4 rectangle under a triangle of height 3	rectangle + triangle	33
a rectangle 14×10 with a semicircle of radius 7 on top	rectangle + semicircle	217
a running-track shape: 80×70 rectangle with two semicircular ends of radius 35	rectangle + circle	9450

WRITE THE AREA OF EACH PIECE ON THE SKETCH

Almost every lost mark in compound-area questions is a piece counted twice or left out — not an arithmetic error.

Subtracting

SHAPE	WORKING	AREA
a 20×15 plot with a 4×3 shed	$300 - 12$	288
a square of side 14 with a disc of radius 7 cut out	$196 - 154$	42
a 1 m path round a 5×4 m plot	$42 - 20$	22
a ring between radii 7 and 14	$616 - 154$	462

SUBTRACTING IS OFTEN QUICKER THAN CUTTING

A path round a plot could be split into four rectangles, but subtracting the inner area from the outer is one line of working instead of five.

Perimeter by tracing

Put a finger at one point of the boundary and travel round, writing down every edge you cross. Internal cuts are not part of the perimeter.

SHAPE	BOUNDARY PIECES	PERIMETER
rectangle 14×10 with a semicircle of radius 7 on top	$10 + 14 + 10 + 22$	56
a square of side 14 with a disc cut out	four sides + the circle	$56 + 44 = 100$
an L-shape 10×8 less a 4×3 corner	six sides	36

THE CUT EDGE IS NOT ON THE BOUNDARY

Where the semicircle meets the rectangle, that 14 cm line is inside the shape. It counts for neither perimeter piece — but the bottom 14 cm does.

Mixed problems

PROBLEM	AREA	PERIMETER
a semicircle of radius 7 on a 14×10 rectangle	217 cm^2	56 cm
a 20×12 plot with a quadrant lawn of radius 7	$240 - 38.5 = 201.5\text{ m}^2$	—
a track: 80×70 rectangle plus two semicircular ends of radius 35	9450 m^2	380 m

TWO ANSWERS, TWO UNITS

An area is in m^2 and a perimeter in m . Writing the units is the last check that the right question has been answered.

02 Worked examples

EXAMPLE 1 A semicircle of radius 7 cm sits on top of a rectangle 14 cm by 10 cm . Find the area.

- ① Rectangle: $14 \cdot 10 = 140\text{ cm}^2$.
- ② Semicircle: $\frac{1}{2} \cdot \frac{22}{7} \cdot 49 = 77\text{ cm}^2$.
- ③ $140 + 77 = 217\text{ cm}^2$.

ANSWER

217 cm^2

EXAMPLE 2 Find the perimeter of the same shape.

- ① Trace the boundary: two sides of 10 , the bottom of 14 , then the curve.
- ② The curve is $\pi r = 22\text{ cm}$.
- ③ $10 + 14 + 10 + 22 = 56\text{ cm}$.
The top 14 is internal.

ANSWER

56 cm

EXAMPLE 3 A square of side 14 cm has a disc of radius 7 cm cut out of it. Find the area left.

- ① Square: 196 cm^2 .
- ② Disc: 154 cm^2 .
- ③ $196 - 154 = 42\text{ cm}^2$.
The four corner pieces.

ANSWER

42 cm^2

03 Interactive model

Ask for area and perimeter of the same compound shape in every exercise; separating the two is the whole difficulty of the topic.

Add, subtract, or trace?

An L-shape 10×8 less a 4×3 corner has area:

A 14×10 rectangle with a semicircle of radius 7 on top has area:

Its perimeter is:

A square of side 14 less a disc of radius 7 has area:

42

154

196

350

Its perimeter is:

56

44

100

12

A 1 m path round a 5×4 plot has area:

20

22

42

9

An internal cut edge counts in:

the area

the perimeter

both

neither

Areas are written in:

m

m²

m³

no units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Compound figure	Murakkab shakl	Составная фигура
To decompose	Bo'laklarga ajratish	Разбить на части
To subtract	Ayirmoq	Вычесть
Boundary	Chegara	Граница
Internal edge	Ichki chegara	Внутренняя линия
Rectangle	To'g'ri to'rtburchak	Прямоугольник
Semicircle	Yarim doira	Полукруг
Sum of the parts	Qismlar yig'indisi	Сумма частей

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A compound area is found by:

measuring

cutting into known pieces

guessing

tracing

A compound perimeter is found by:

adding all the pieces' perimeters

tracing the boundary

subtracting

halving

A 20×15 plot less a 4×3 shed:

288

300

312

12

A ring between radii 7 and 14:

154

462

616

770

Where a semicircle meets a rectangle, the shared edge:

counts once

counts twice

does not count

is halved

The best first step is:

calculate

sketch and label the pieces

estimate

measure

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. An L-shape 10×8 less 4×3

ANSWER 68

2. A house: 6×4 plus a triangle of height 3

ANSWER 33

3. A 20×15 plot less a 4×3 shed

ANSWER 288

4. A 1 m path round a 5×4 plot

ANSWER $22 m^2$

5. A square of side 14 less a disc of radius 7

ANSWER $42 cm^2$

6. A ring between radii 7 and 14

ANSWER $462 cm^2$

7. Units for an area

ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. A 14×10 rectangle plus a semicircle of radius 7

ANSWER 217 cm^2

2. Its perimeter

ANSWER 56 cm

3. The square-less-disc shape: its perimeter

ANSWER 100 cm

4. The L-shape: its perimeter

ANSWER 36

5. A track: 80×70 plus ends of radius 35: the area

ANSWER 9450 m^2

6. And its perimeter

ANSWER 380 m

7. A 20×12 plot less a quadrant of radius 7

ANSWER 201.5 m^2

Hard · stretch and reason

7 PROBLEMS

1. A 2 m path round a 10×6 m garden

ANSWER 80 m^2

2. A 28 cm square with quadrants of radius 14 cut from two corners

ANSWER 476 cm^2

3. A shape of two semicircles of radius 7 on opposite sides of a 14×20 rectangle: the area

ANSWER 434 cm^2

4. Its perimeter

ANSWER 84 cm

5. A disc of radius 14 with a 14 cm square hole

ANSWER 420 cm^2

6. Why is subtracting often better than cutting?

ANSWER One line of working instead of several

7. A 60° sector of radius 21 cut from a 21×21 square

ANSWER 210 cm^2

07 Homework

Homework — 5 tasks

Sketch, label every piece, then answer area and perimeter separately.

- Find the area of an L-shape 12×9 with a 5×4 corner removed, and its perimeter.
- A semicircle of radius 10 cm sits on a 20×15 cm rectangle. Find the area ($\pi = 3.14$).
- Find the perimeter of the same shape.
- A 1.5 m path runs round a 12 m by 8 m lawn. Find the area of the path.
- A square of side 28 cm has a disc of radius 14 cm cut out. Find the area remaining.